RAJALAKSHMI ENGINEERING COLLEGE

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C2

Started on	Wednesday, 7 October 2026, 10:12 PM
State	Finished
Completed on	Wednesday, 7 October 2026, 10:13 PM
Time taken	25 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Find Duplicate in Array.

Given a read only array of n integers between 1 and n, find one number that repeats.

Input Format:

First Line - Number of elements

n Lines - n Elements

Output Format:

Element x - That is repeated

For example:

Input	Result
5 1 1 2 3 4	1

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>
#include <stdlib.h>

int main() {
    int n;
    // Read the number of elements
    if (scanf("%d", &n) != 1) return 0;

    // Allocate a frequency tracker initialized to 0
    int *seen = (int *)calloc(n + 1, sizeof(int));
    if (seen == NULL) return 1;

    int duplicate = -1;

    // Read n elements
    for (int i = 0; i < n; i++) {
        int val;
        if (scanf("%d", &val) != 1) {
            free(seen);
            return 0;
        }

        // If we have already seen this number, it's the duplicate
        if (seen[val] == 1 && duplicate == -1) {
            duplicate = val;
        }
        seen[val] = 1;
    }

    // Print the duplicate number
    printf("%d", duplicate);

    // Clean up memory
    free(seen);
    return 0;
}

```

	Input	Expected	Got	
✔	11 10 9 7 6 5 1 2 3 8 4 7	7	7	✔
✔	5 1 2 3 4 4	4	4	✔
✔	5 1 1 2 3 4	1	1	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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